

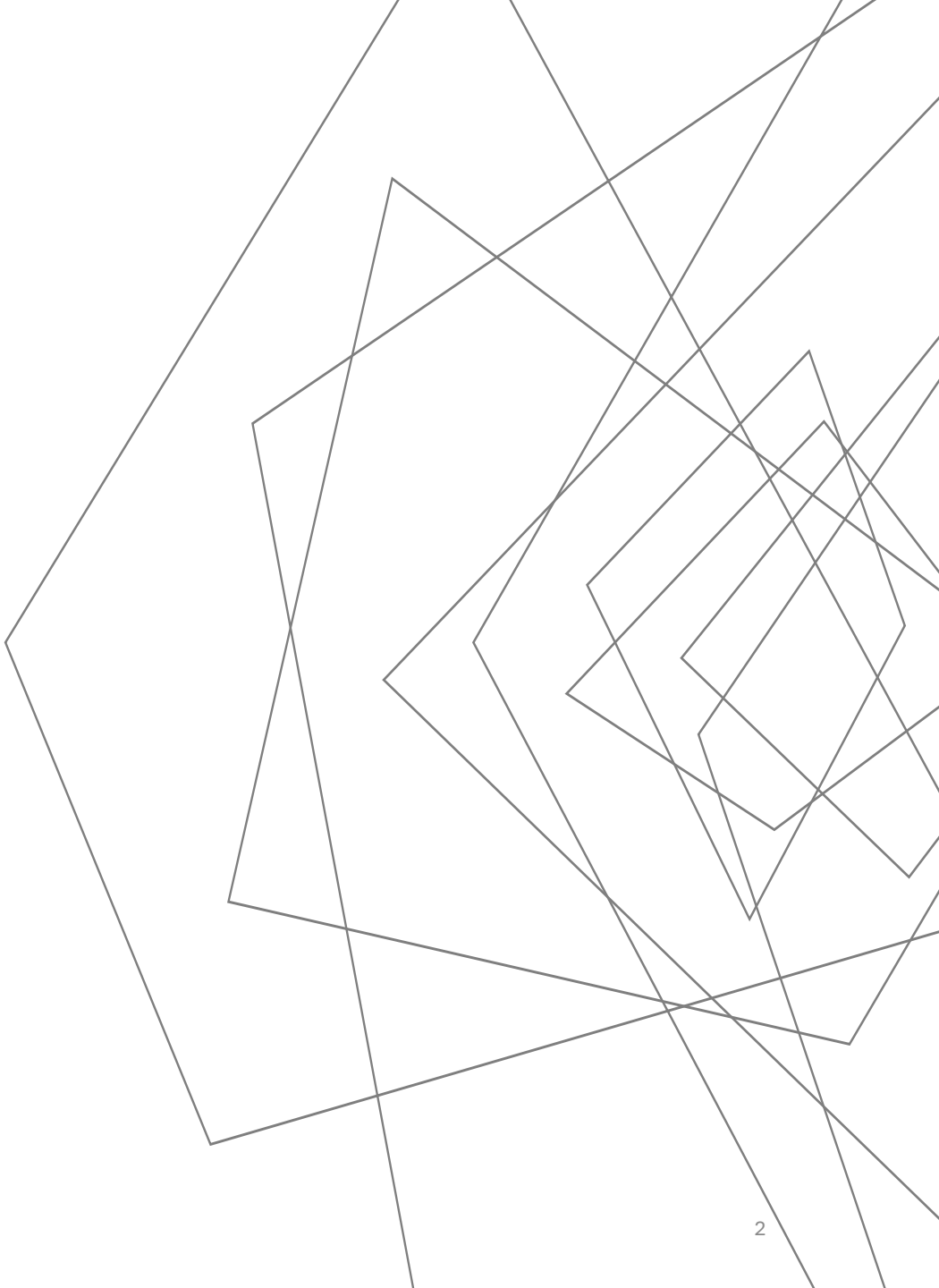


**Hof
University**
University of
Applied Sciences

APPLIED ARTIFICIAL INTELLIGENCE

02 – INTRODUCTION

PROF. DR. LARS ACKERMANN-IGL



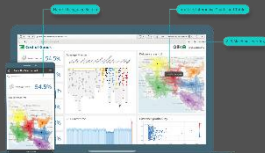
MOTIVATION & USE CASES

- Use cases from different domains and areas
- Abstracting the use cases to general AI problems
- Basic ML terms

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 1



SELF-SERVICE ANALYTICS



Offer employees the possibility for automatic report generation / evaluation

APPLICATION CLASSIFICATION



Automatic processing and preselection of applications

ASSESSMENT AUTOMATION



Automation of the assessment process, e.g., based on automatic generation of questionnaire or automatic evaluation

POSITION MATCHMAKING



Automatic matches / proposals for positions based on known employee skillset

INFORMATION RETRIEVAL



Central platform for information retrieval, e.g. possibility to crawl unstructured information

Potential applications in every domain in an organization

Administration & HR

Marketing & Customer Relations

Sales & Procurement

Warehousing & Intralogistics

Maintenance & Services

Innovation management

Research & Development

Supply Chain Management

Resource Planning & Manufacturing



Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 3

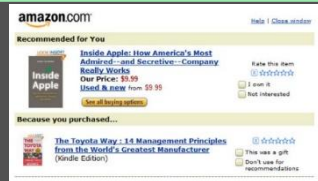


TARGET MARKETING



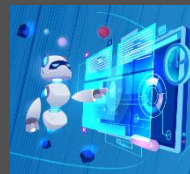
Target based marketing with respect to known information about individuals

RECOMMENDATION SYSTEMS



Recommendation of similar products based on customer behavior / sales data

AUTOMATIC CAMPAIGNS



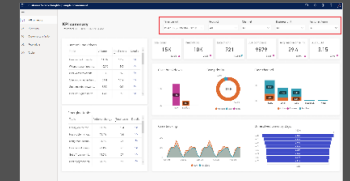
Generation / publishing of e.g., social media content e.g., based on competitor activity analysis

CAMPAIGN ANALYSIS



Automatic analysis of the impact of e.g. a social media campaign based on KPIs

SATISFACTION ANALYSIS / REPORTING



Automatic reporting of customer satisfactory in e.g., a dashboard application

Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 4

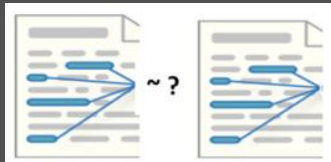


PRODUCT OPTIMIZATION / USAGE ANALYSIS



Optimize product features, e.g. remove not needed features based on usability data

PATENT RESEARCH



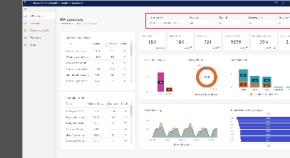
Automatic research of potentially relevant patents used in product development

PROCESS SIMULATION



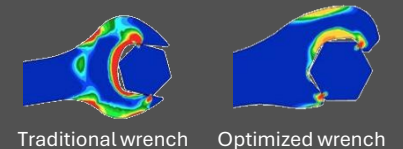
Simulation of processes in order to virtualize product development / fast prototype

PROCESS DATA VISUALIZATION



Visualize product / process data in order to get insights

PROCESS OPTIMIZATION



Optimize the product / process, e.g. minimize materials, optimize design

Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 5

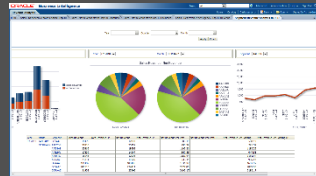


MARKET AREA ANALYSIS



Automatic analysis / reporting of the potential of a market area

SALES CHANNEL ANALYSIS



Automatic analysis of the performance of a sales channel

PREDICTIVE PROCUREMENT



Prediction of e.g. the price of raw materials and data-driven procurement decision

DEMAND ANALYSIS / FORECAST



Automatic analysis / forecast of demand, e.g. demand of material / resources

DYNAMIC PRICING



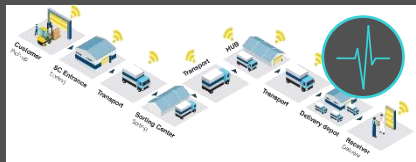
Automatic pricing based on observation of the environment

Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 6

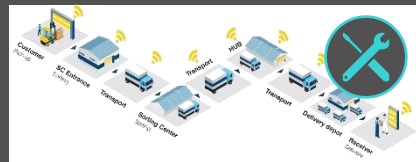


SUPPLY CHAIN HEALTH ANALYSIS



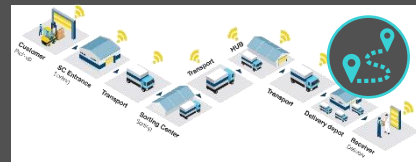
Automatic analysis of the health of the supply chain, e.g. identifying bottlenecks

SUPPLY CHAIN OPTIMIZATION



Optimization of the supply chain, by e.g. eliminating bottlenecks or not needed partners

SUPPLY CHAIN TRACKING



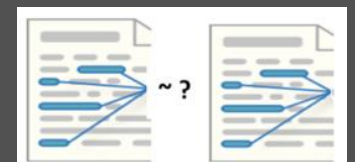
Track products over the complete supply chain, from raw material to end-customer

DEMAND ANALYSIS / FORECAST



Part demand forecast with automatic adoption of the supply chain

CONTRACT ANALYSIS



Analysis and standardization of contracts which are existing with all supply chain partners

Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 7

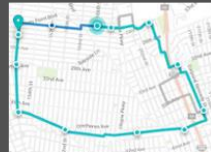


INVENTORY OPTIMIZATION



Optimize the inventory management, e.g. by minimizing stock

ROUTE OPTIMIZATION



Optimization of routes, e.g. for forklifts or autonomous vehicles

AUTONOMOUS VEHICLES



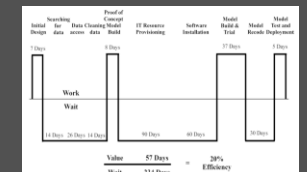
Usage of autonomous vehicles within intralogistics

FLEET MANAGEMENT



Management of intralogistics fleet, e.g. optimized service plan / schedules

VALUE STREAM OPTIMIZATION



Identify waste (e.g. overproduction / waiting times / ...) and maximize the value stream

Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 8



MONITORING



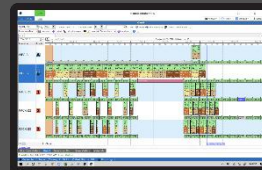
Monitor production machines (e.g. acoustic monitoring) and early identification of potential failures

ASSEMBLY OPTIMIZATION



Analysis and optimization of product assembly in order to maximize production speed, minimize material,...

OPTIMIZED SCHEDULING



Optimize production scheduling in order to minimize the downtime of a machine

PREDICTIVE QUALITY



Automatic analysis of the manufactured parts and automatic identification of good/bad parts

AUTOMATED ROOT-CAUSE-ANALYSIS



Automatic root-cause analysis when errors occur during production

Potential applications in every domain in an organization

MOTIVATION: APPLICATION FIELDS / USE CASES FOR AI – 9



PREDICTIVE MAINTENANCE



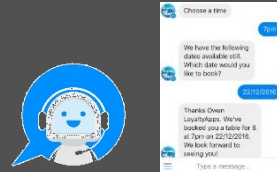
Automatic prediction of the next necessary maintenance tasks

REPAIR COST PREDICTION



Automatic analysis of necessary maintenance and identification of maintenance costs

AUTOMATED 1ST LEVEL SUPPORT



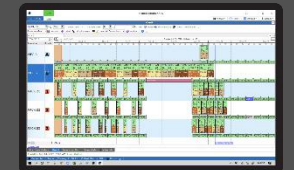
Automation of first level support in order to offer a 24/7 helpline, e.g. using a chatbot solution

PRODUCT LIFETIME PREDICTION



Automatic prediction of the left product lifetime

OPTIMIZED MAINTENANCE SCHED.



Optimized scheduling for required maintenance tasks, e.g. minimize downtime

Potential applications in every domain in an organization



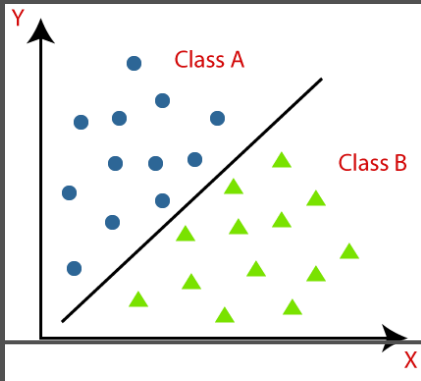
AI/MACHINE LEARNING BASICS

- Basic terms in AI and machine learning
 - AI problem categories
 - Types of AI and machine learning approaches
- A machine learning process
- Evaluation (metrics, plotting)

AI PROBLEM CATEGORIES

Not part of this lecture

CLASSIFICATION



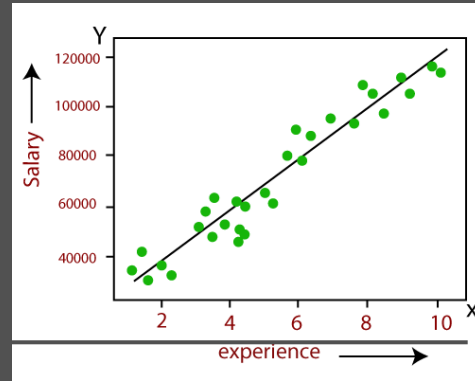
Short description:

- Classification tasks aim to predict a discrete value (label)

Example methods:

- Naive Bayes
- Support Vector Machines
- Decision Trees
- Neural Networks

REGRESSION



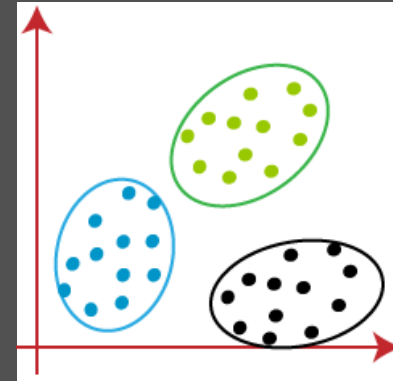
Short description:

- Regression tasks aim to predict a continuous label (and/or there is a native ordering in the labels)

Example methods:

- Linear Regression
- Non-linear Regression
- Neural Networks

CLUSTERING



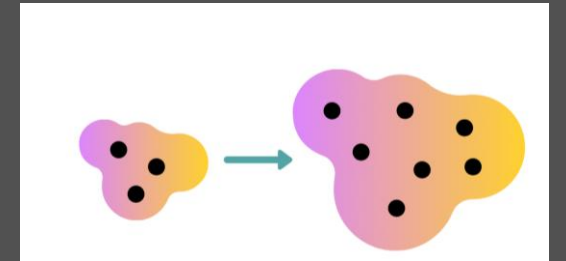
Short description:

- Clustering tasks aim to group the dataset into partitions

Example methods:

- k-Means
- DBSCAN
- Nearest Neighbors

GENERATION



Short description:

- Generation tasks aim to generate synthetical samples for a given prediction

Example methods:

- Variational Autoencoders
- Generative Adversarial Networks
- Transformer Networks

Classification, Clustering and Regression are the core problem categories in artificial intelligence.

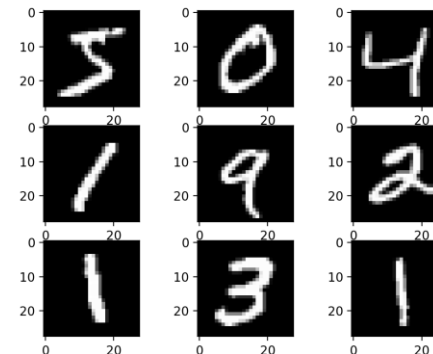
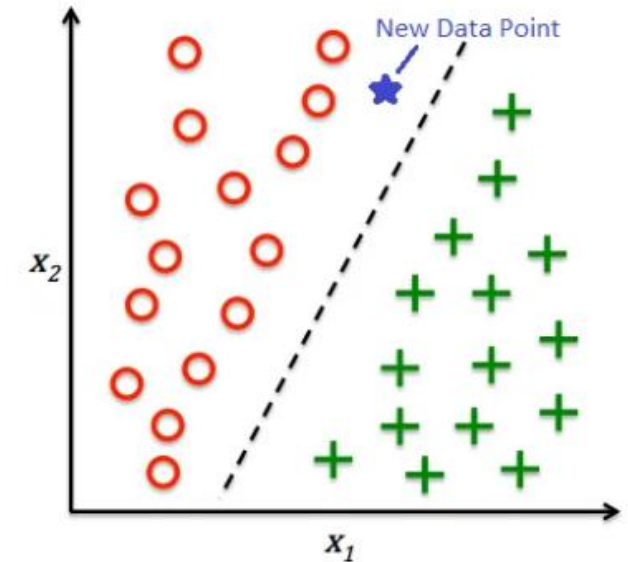
CLASSIFICATION – A CLOSER LOOK

Principle

- Approach: Creation of a model that describes the relationship between characteristics and class membership
- Goal: Prediction of class membership for new data points.

Variants

- Binary classification
 - Two classes
 - Example: Email spam vs. no spam
- Multi-class classification
 - Multiple classes
 - Example: Recognition of handwritten digits (0-9)



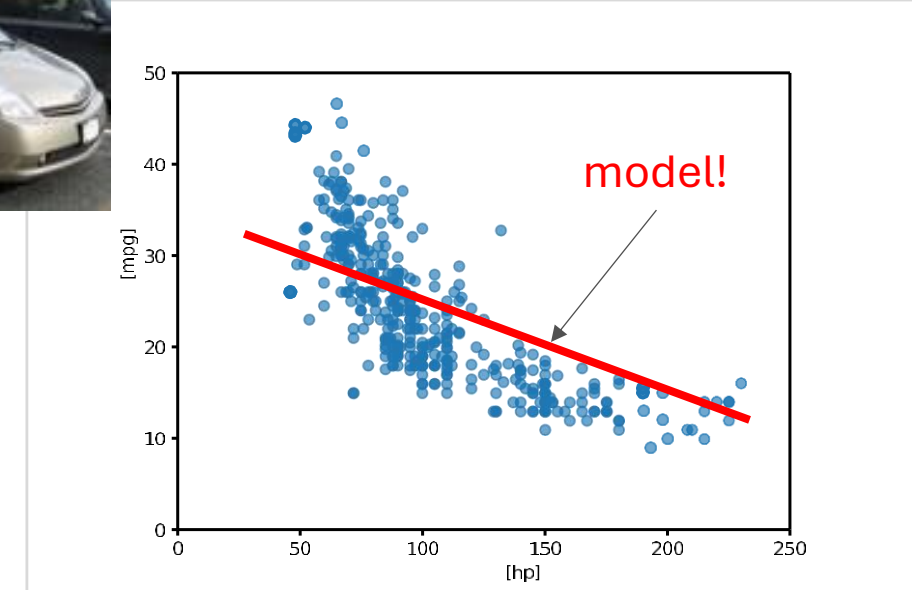
REGRESSION – A CLOSER LOOK

Principle

- Approach: Creation of a model that describes the relationship between characteristics x_i and a numerical (continuous) variable y .
- Goal: Find out the relationship between x_i and y and predict y given an (unseen) x .

Example: Estimating a car's fuel consumption

- Real-life dataset: Auto MPG data set from UCI ML Repository
<https://archive.ics.uci.edu/ml/datasets/auto+mpg>
398 cars (392 with full features)
- 8 characteristics (consumption, cylinder, weight, etc.)
- Simplification for our lecture (see right side):
 - Independent variable: horse power (hp)
 - Dependent variables: consumption (mpg)



KAHOOT TIME!

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BASIC TERMS OF MACHINE LEARNING (ML) – 1

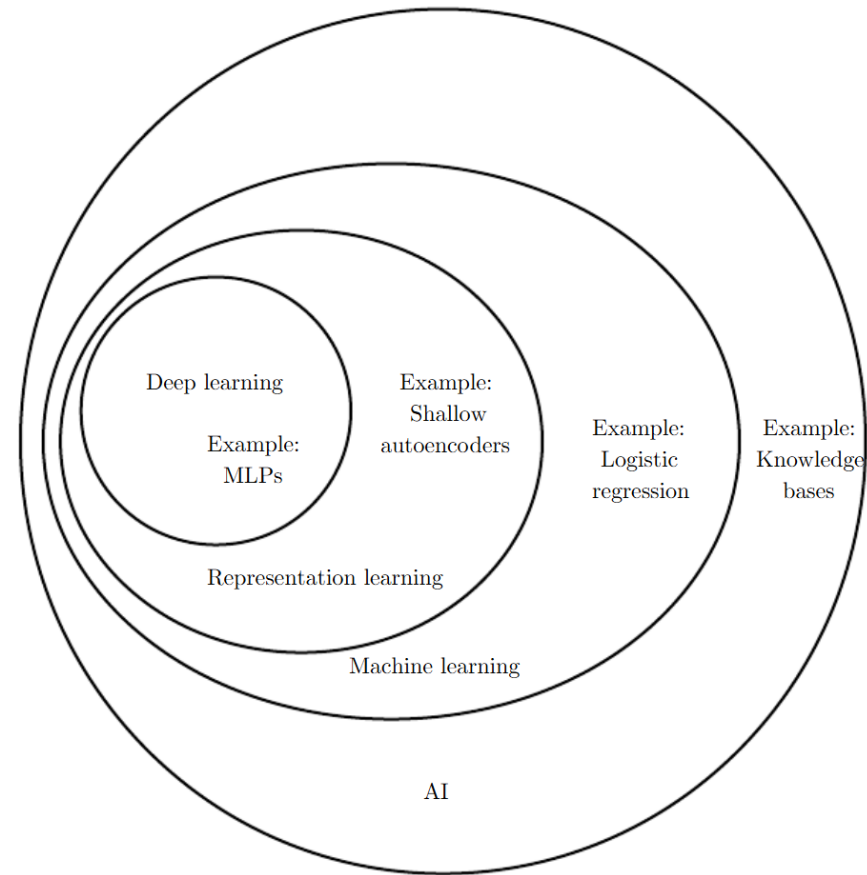


Figure 1.4: A Venn diagram showing how deep learning is a kind of representation learning, which is in turn a kind of machine learning, which is used for many but not all approaches to AI. Each section of the Venn diagram includes an example of an AI technology.

[Goodfellow, p. 9]

BASIC TERMS OF MACHINE LEARNING (ML) – 2

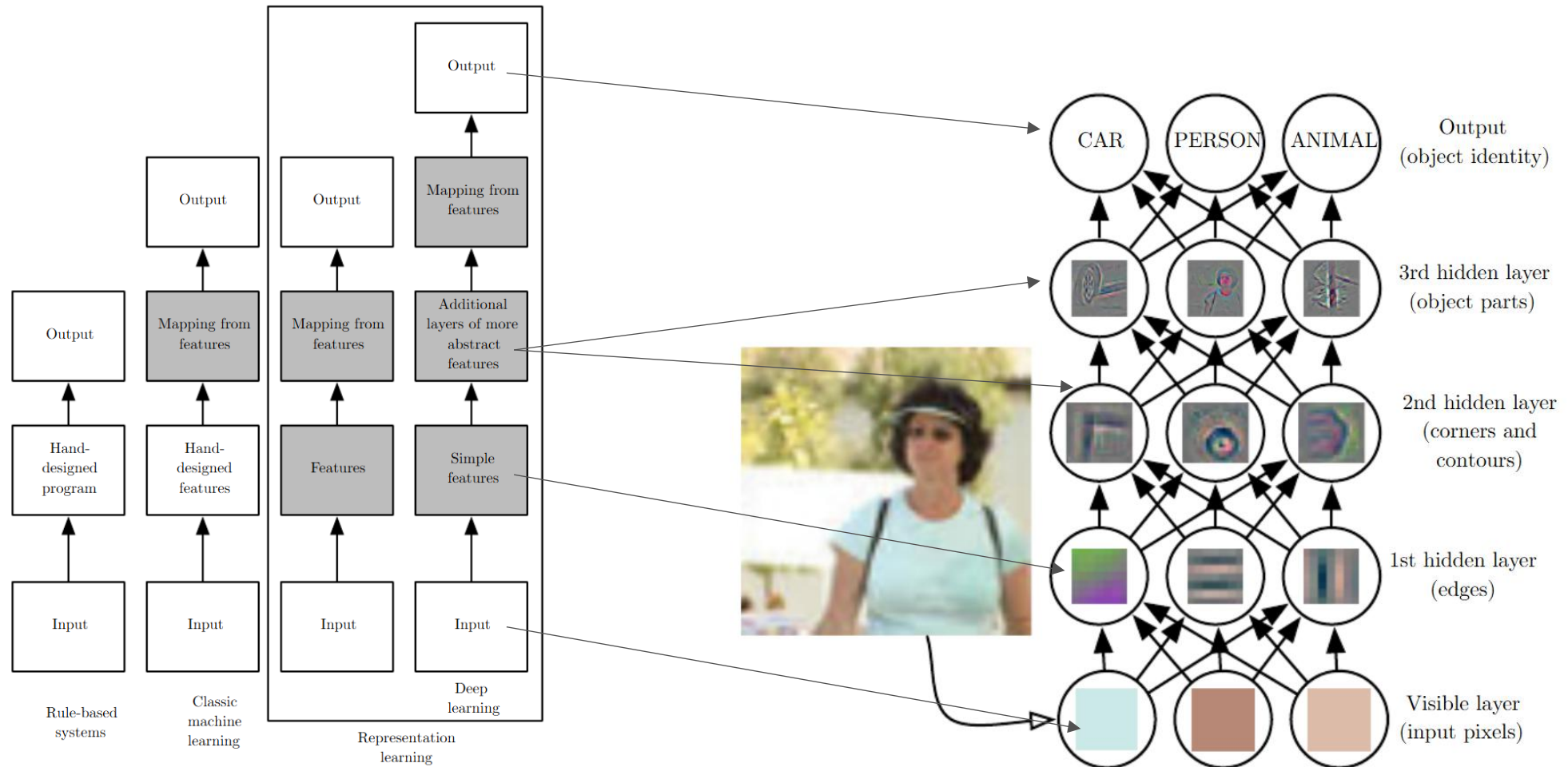


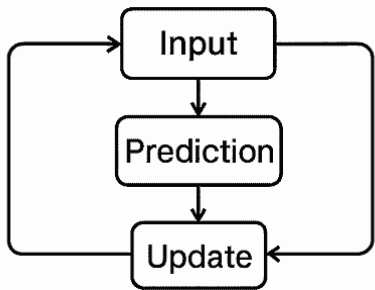
Figure 1.5: Flowcharts showing how the different parts of an AI system relate to each other within different AI disciplines. Shaded boxes indicate components that are able to learn from data.

[Goodfellow, p. 10]

[Goodfellow, p. 6]

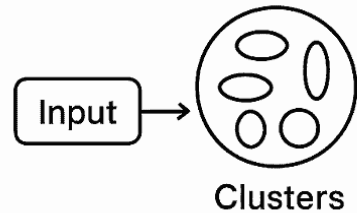
BASIC TERMS OF MACHINE LEARNING (ML) – 3

SUPERVISED LEARNING



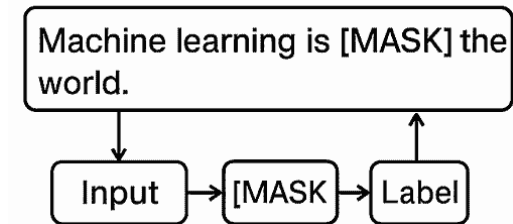
Supervised learning methods use **labeled** data (e.g. images of cats/dogs) as input and train a model in order to **minimize** the **classification error**. Labeled data is typically **split** into a **train** and **test** (and evaluation) set.

UNSUPERVISED LEARNING



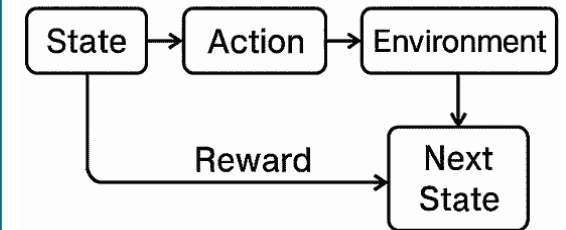
Unsupervised learning methods use **unlabeled** data as input and aim to **find groups / clusters** of similar data elements (with respect to a defined **distance metric**). Unsupervised methods can only detect **similarity** (no classes).

SELF-SUPERVISED LEARNING



Self-supervised learning methods use **unlabeled data** but create **pseudo-labels** from the data itself (e.g., predicting masked parts). The goal is to learn useful **representations** that can be used for downstream tasks.

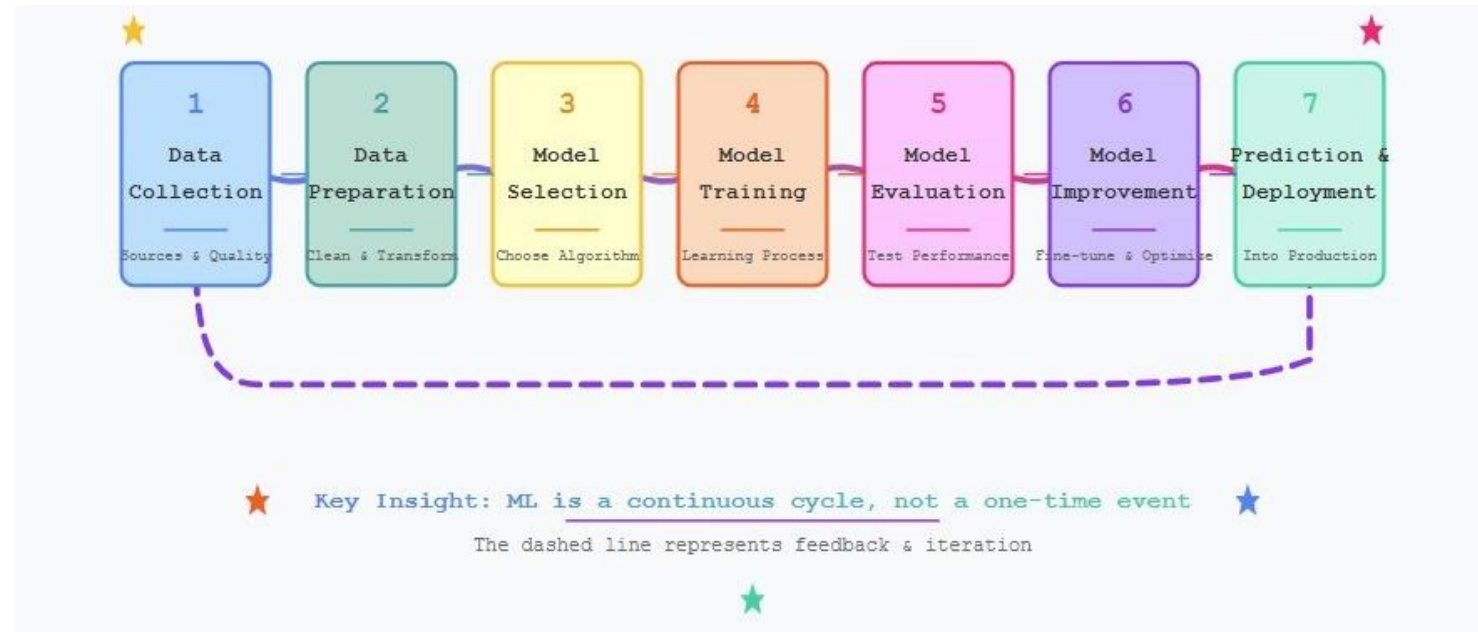
REINFORCEMENT LEARNING



Reinforcement learning methods are e.g. often used within robotics / autonomous driving. These models take **states** of the environment and possible **actions** as input and calculate an action as output, which has the **highest reward**.

In this course, we will focus primarily on supervised learning and cover unsupervised learning and reinforcement learning in less detail.

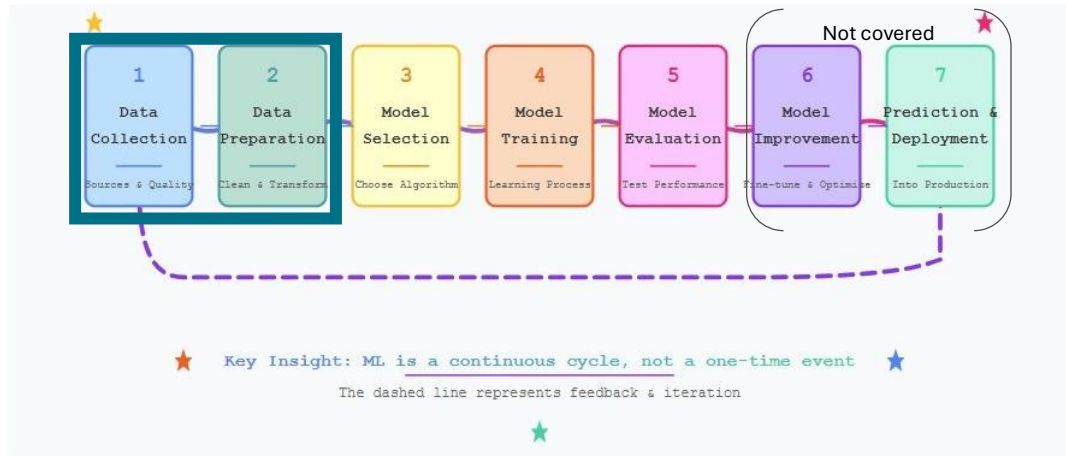
A MACHINE LEARNING PROCESS – OVERVIEW



HOW A (SIMPLIFIED) MACHINE LEARNING PROCESS LOOKS LIKE

(NOTE: IT IS TYPICALLY NOT THAT LINEAR!)

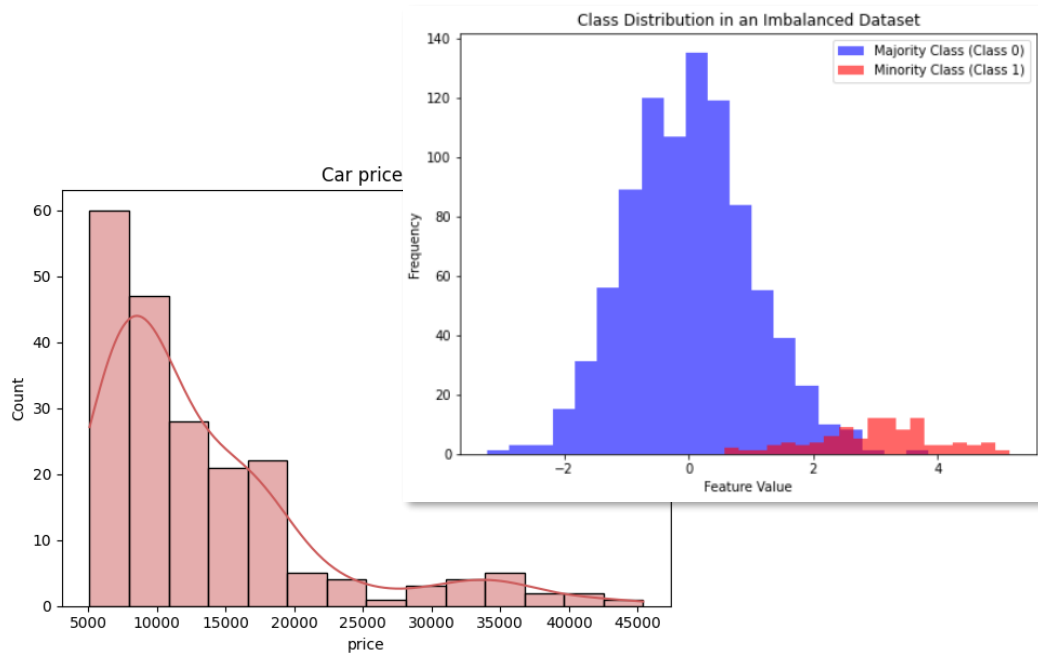
A MACHINE LEARNING PROCESS – THE DATA



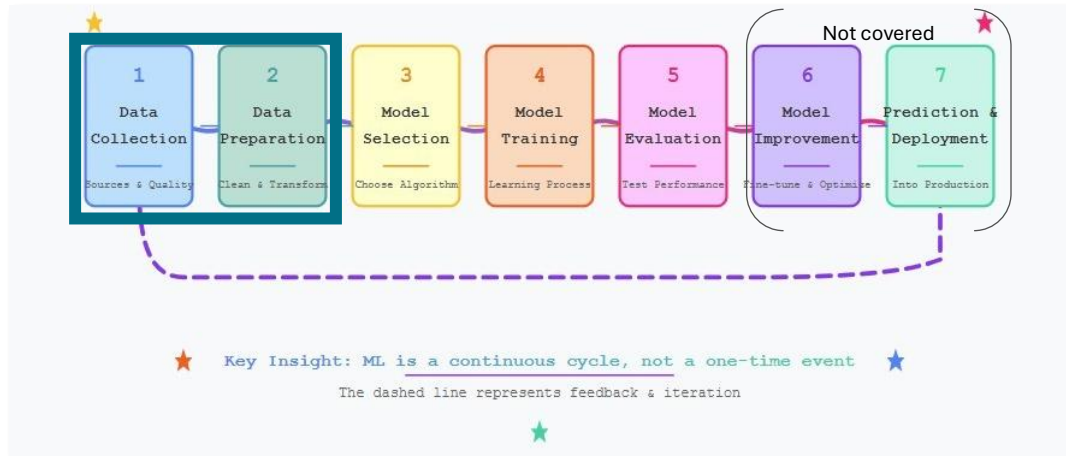
Data Collection and preparation

- Integrate data (merge from multiple sources -> “ETL process”)
- Explore
 - Label distribution (for classification and regression problems)
 - Feature distribution (for clustering)
- Data Cleaning / Cleansing
 - Outlier
 - Missing values
- Data Transformation
 - Standardization / Normalization
 - Type change

Let's talk about this a bit more in detail!



A MACHINE LEARNING PROCESS – THE DATA



Standardization

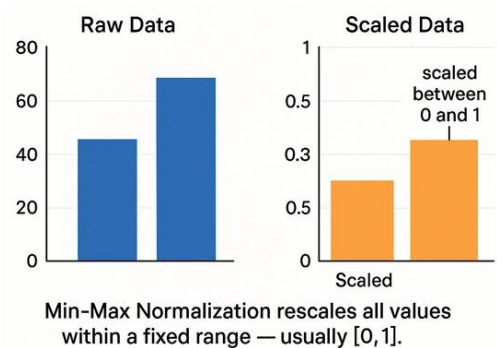
What it does: Align feature scales

Why it is useful: Prevents dominance of large-magnitude variables; improve stability & convergence

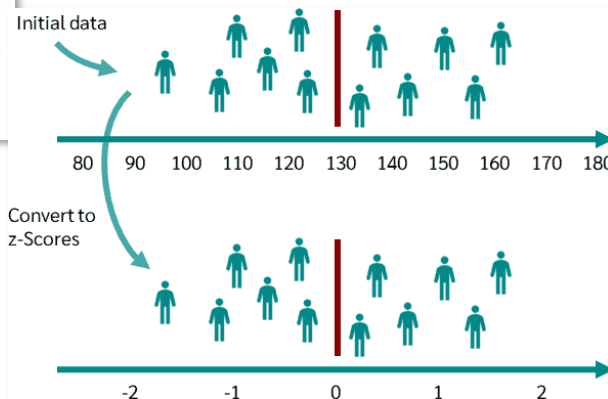
Two examples:

- **Min-max normalization**
(maps data to fixed range [0,1] of same distribution shape)

min-max normalization



z-score normalization



$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

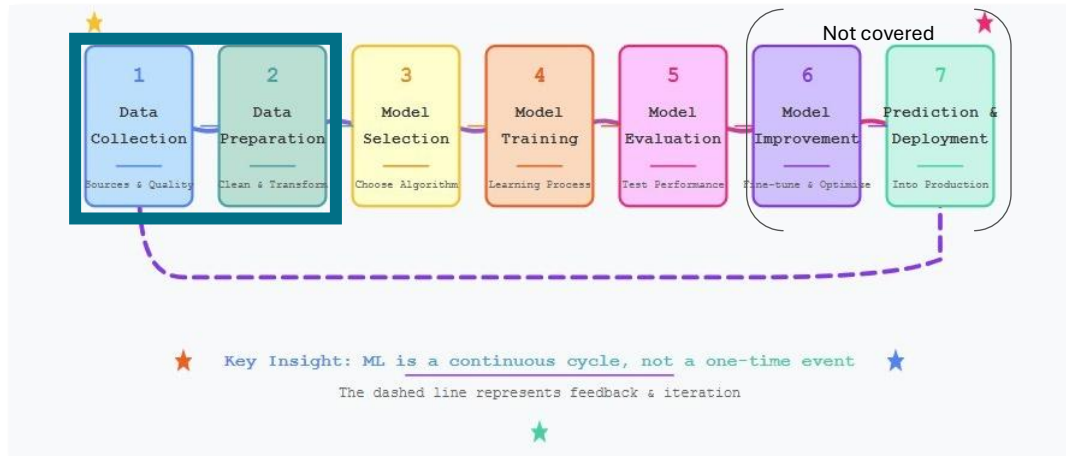
-> use when a fixed numeric range is mandatory

- **Z-score normalization**
(centers data around 0 with standard deviation of 1; focus on relative distance from the mean value):

$$x' = \frac{x - \mu}{\sigma}$$

-> use when features have different scales but no extreme outliers

A MACHINE LEARNING PROCESS – THE DATA



Ordinal encoding

Original Encoding	Ordinal Encoding
Poor	1
Good	2
Very Good	3
Excellent	4

One-hot vector encoding

Color	One-hot encoding	d1	d2	d3
Red	1	1	0	0
Green	1	0	1	0
Blue	1	0	0	1

Type change (of categorical variables)

What it does: Maps feature values or classes to different data type

Why it is useful: Many ML algorithms require numerical input

Two examples:

- **Ordinal encoding**
(maps data to subsequent integer values)

$$x' = f(c) \text{ with } f(c) \in \{0, 1, 2, \dots, k - 1\}$$

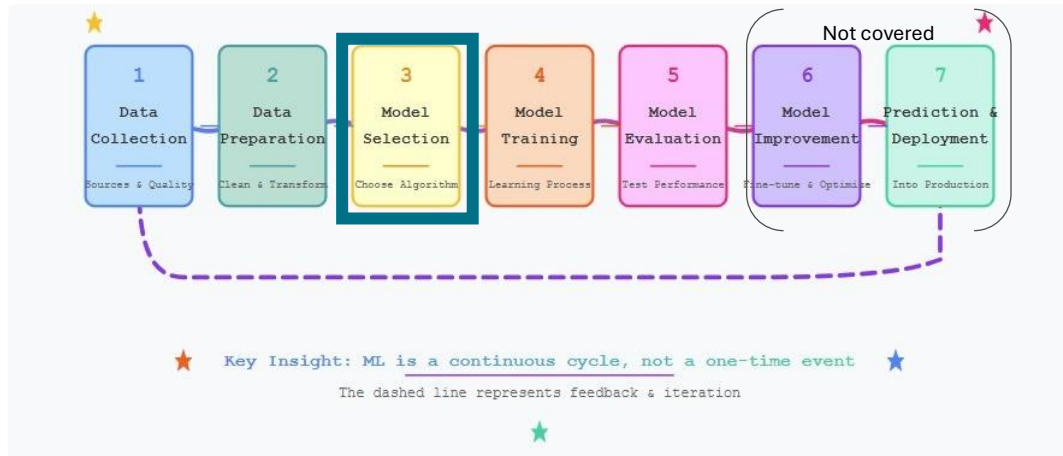
-> use in case the ordering matters

- **One-hot vector encoding**
(represent each category k as a binary vector with exactly one component that is 1 and all others are 0 (treats all k categories as equally distant))

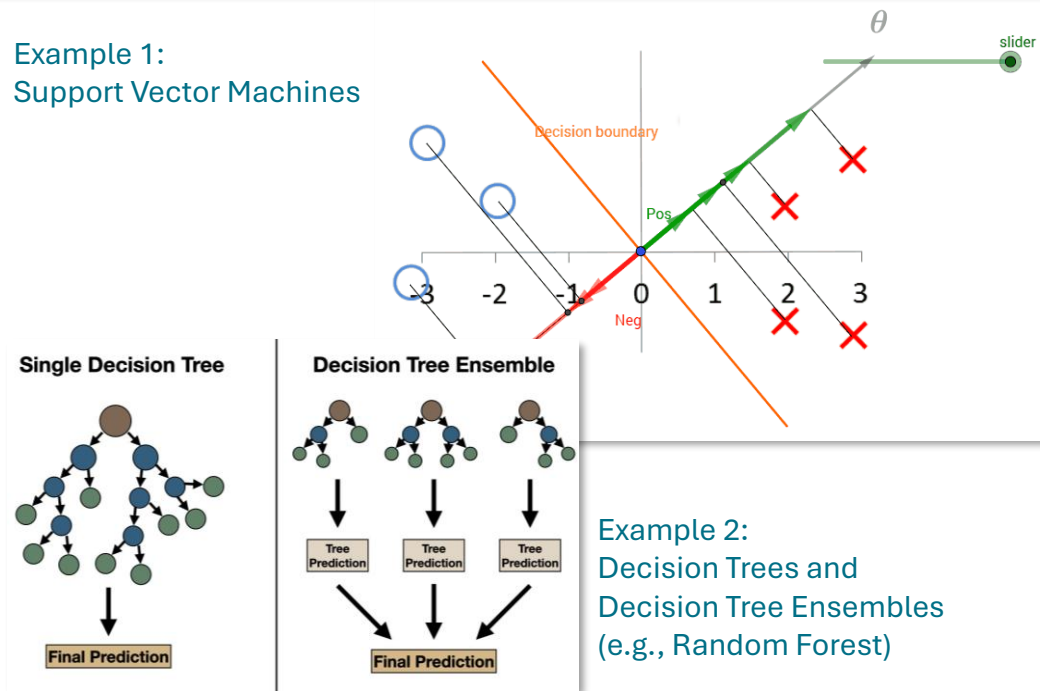
$$x_i' = \begin{cases} 1 & \text{if } i = 1 \\ 0 & \text{else} \end{cases}$$

-> in case classes are independent; ordering does not matter

A MACHINE LEARNING PROCESS – CHOICE OF MODELS

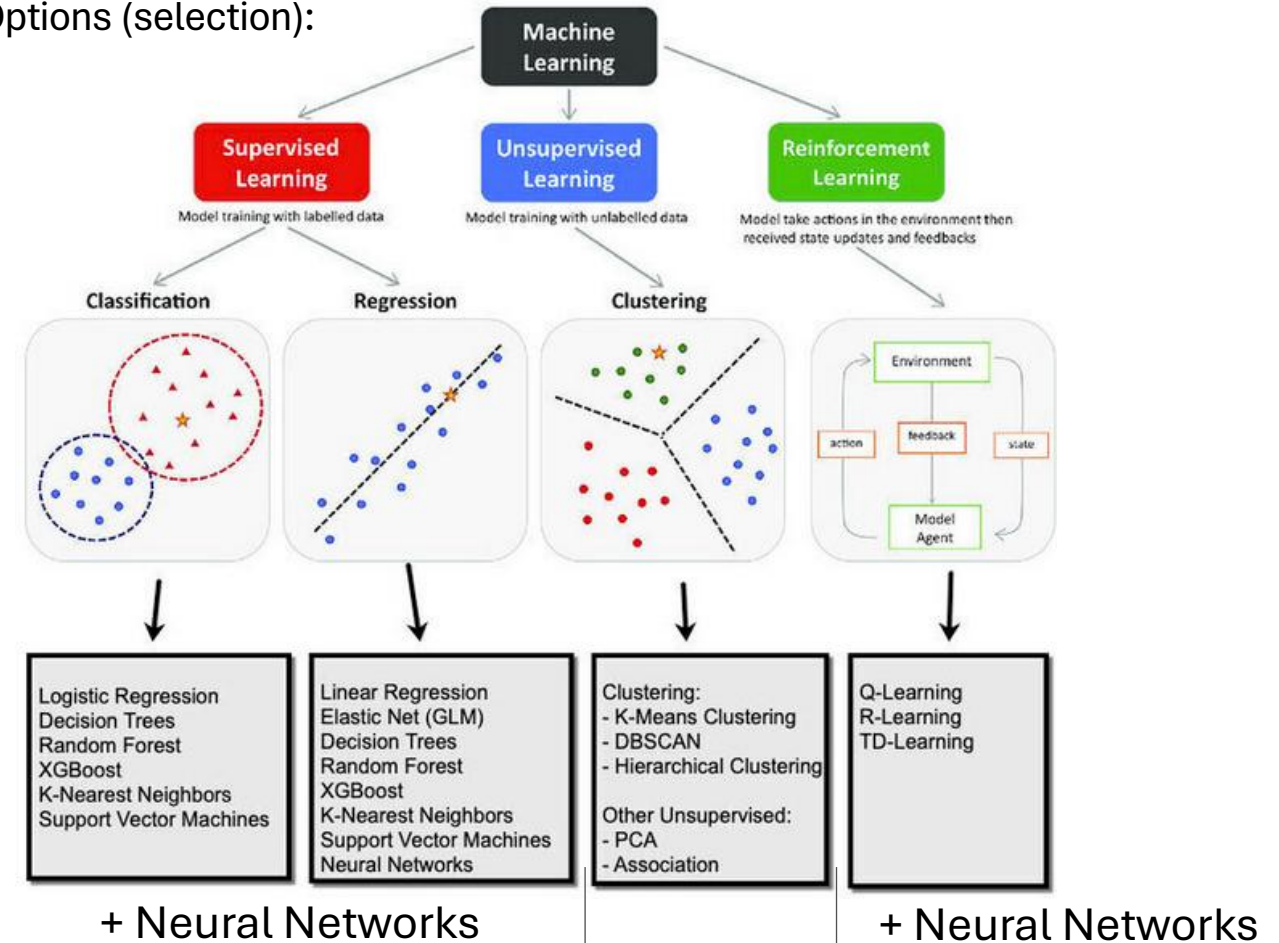


Example 1: Support Vector Machines

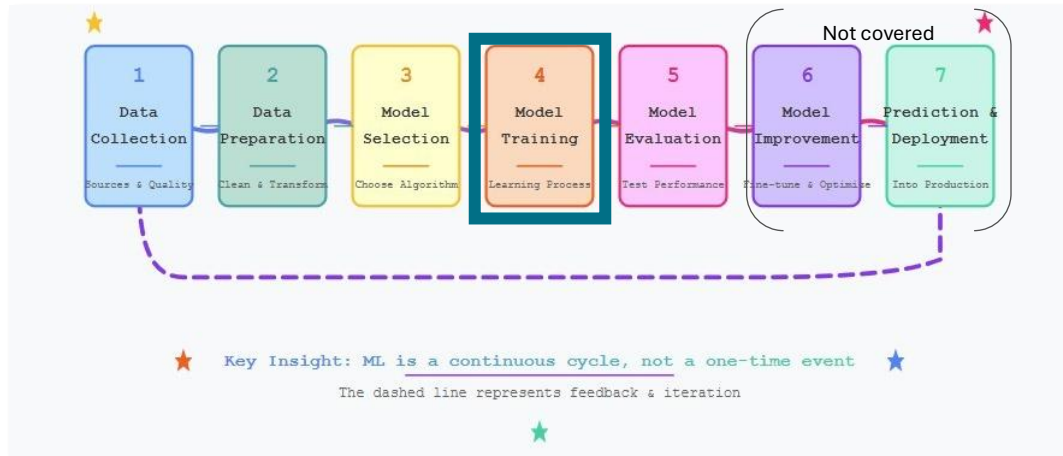


Example 2: Decision Trees and Decision Tree Ensembles (e.g., Random Forest)

- The choice of an appropriate model is a matter of experience and – unfortunately – an empirical problem!
- Options (selection):

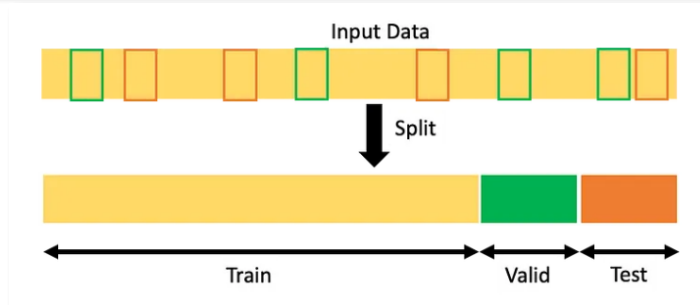
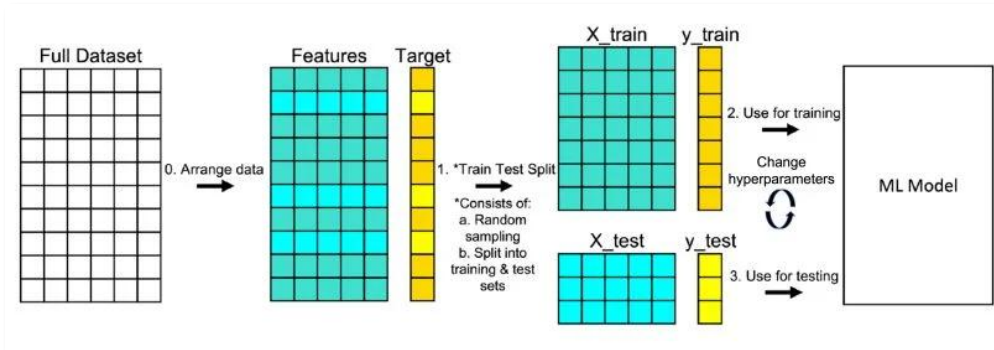


A MACHINE LEARNING PROCESS – THE TRAINING



Train-test splits

- Before training, the dataset is usually split into two parts:
 - Training set** – used to teach the model.
 - Test set** – used to evaluate how well the model performs on new, unseen data.
 - A typical split might be 80% training and 20% test.
- Important rules: $D = D_{train} \cup D_{test}$ and $D_{train} \cap D_{test} = \emptyset$
This ensures we can check the model's real-world performance without cheating.
- In Addition: Use validation splits (during training) for identifying an appropriate hyperparameter configuration (e.g., cross validation for finding the best model)



Setting hyper-parameters

- Hyperparameters: **settings for the learning process; chosen** before training.
Examples: Learning rate (how fast the model updates itself), Number of layers in a neural network, Maximum depth of a decision tree ...
- Choosing hyper-parameters is an optimization problem:
 - Grid search ($m \times n \times o \times \dots$ combinations as a grid)
 - Random Search ($m \times n \times o \times \dots$ random combinations)
 - Tree-Structured Parzen Estimator (TPE) ... models the probability of past trials being “good” or “bad” to pick new hyperparameters likely to be good

Training process

- Model uses training input to compute output
- Model updates internal parameters to reduce some error -> **Wait ... what error??**
-> Next slide

EVALUATION METRICS - OVERVIEW

EVALUATION METRICS

Classification

Clustering

Regression

ACTUAL VALUES	POSITIVE	NEGATIVE		
	TP	FN		
POSITIVE	TP	FN		
NEGATIVE	FP	TN		

$$Precision = \frac{TP}{TP + FP} \quad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

		Predicted			
		Animal	Not animal		
Actual	Model 1	Animal	Not animal		
	Animal	 			
	Not animal	 			

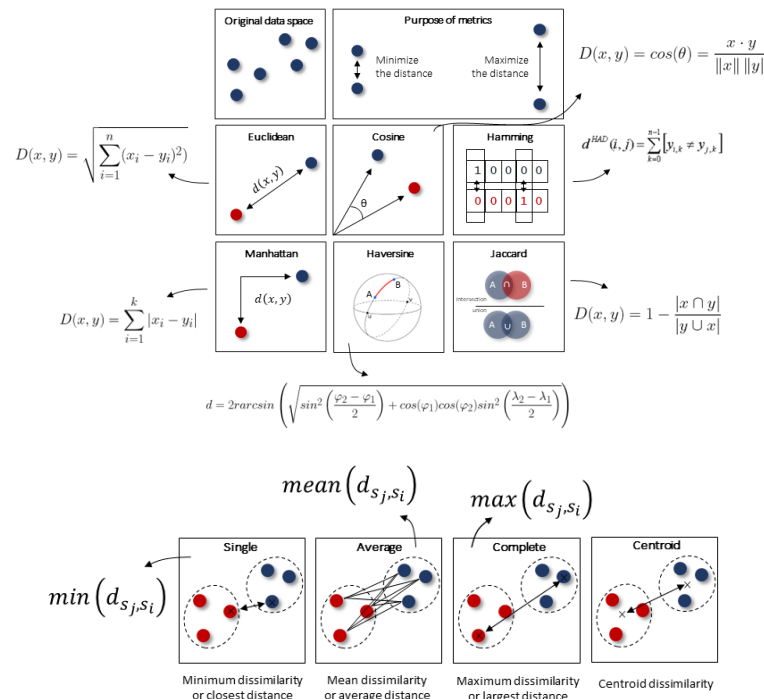
True Positives	3
True Negatives	0
False Positives	3
False Negatives	0

Accuracy	50%	$\frac{3+0}{3+0+0+3}$
Precision	50%	$\frac{3}{3+3}$
Recall	100%	$\frac{3}{3+0}$
F1 score	67%	$2 \cdot \frac{0.5 \cdot 1}{0.5 + 1}$

		Predicted			
		Animal	Not animal		
Actual	Model 2	Animal	Not animal		
	Animal	 			
	Not animal		 		

True Positives	2
True Negatives	3
False Positives	0
False Negatives	1

Accuracy	83%	$\frac{2+3}{2+3+0+0}$
Precision	100%	$\frac{2}{2+0}$
Recall	67%	$\frac{2}{2+1}$
F1 score	80%	$2 \cdot \frac{1 \cdot 0.67}{1 + 0.67}$

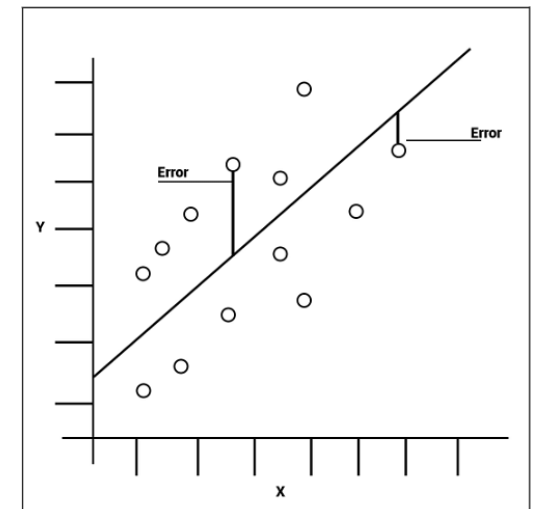


Mean squared error $MSE = \frac{1}{n} \sum_{t=1}^n e_t^2$

Root mean squared error $RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}$

Mean absolute error $MAE = \frac{1}{n} \sum_{t=1}^n |e_t|$

Mean absolute percentage error $MAPE = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{e_t}{y_t} \right|$



EVALUATING CLASSIFICATION MODELS – A CLOSER LOOK

Confusion matrices

Answers the following four questions:

- How many samples did the model correctly predict as class A? [TRUE POSITIVES]
- How many samples did the model correctly predict as not being class A [TRUE NEGATIVES]
- How many samples did the model incorrectly predict as class A? [FALSE POSITIVES] **False alarm!**
- How many samples did the model incorrectly predict as not being class A? [FALSE NEGATIVES] **Underestimation!**

		ACTUAL VALUES	
		POSITIVE	NEGATIVE
ACTUAL VALUES	POSITIVE	TP	FN
	NEGATIVE	FP	TN

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Which one is better?

		MODEL 1 Predicted	
		Animal	Not animal
Actual	Animal	 	
	Not animal	 	

True Positives	2
True Negatives	3
False Positives	0
False Negatives	1

Accuracy	83%	$\frac{2+3}{2+3+1+0}$
Precision	100%	$\frac{2}{2+0}$
Recall	67%	$\frac{2}{2+1}$
F1 score	80%	$2 \cdot \frac{1 \cdot 0.67}{1 + 0.67}$

		MODEL 2 Predicted	
		Animal	Not animal
Actual	Animal	 	
	Not animal	 	

True Positives	3
True Negatives	0
False Positives	3
False Negatives	0

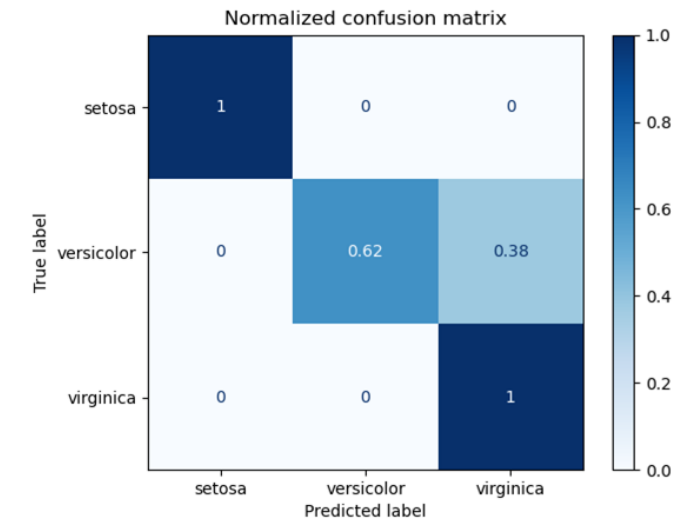
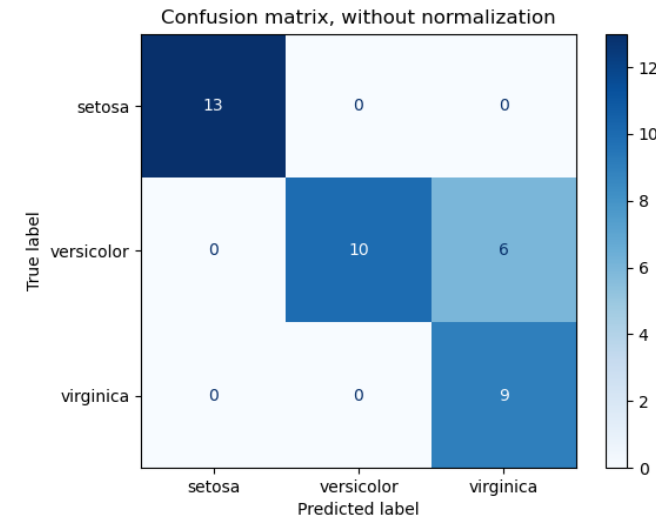
Accuracy	50%	$\frac{3+0}{3+0+0+3}$
Precision	50%	$\frac{3}{3+3}$
Recall	100%	$\frac{3}{3+0}$
F1 score	67%	$2 \cdot \frac{0.5 \cdot 1}{0.5 + 1}$

EVALUATING CLASSIFICATION MODELS – A CLOSER LOOK

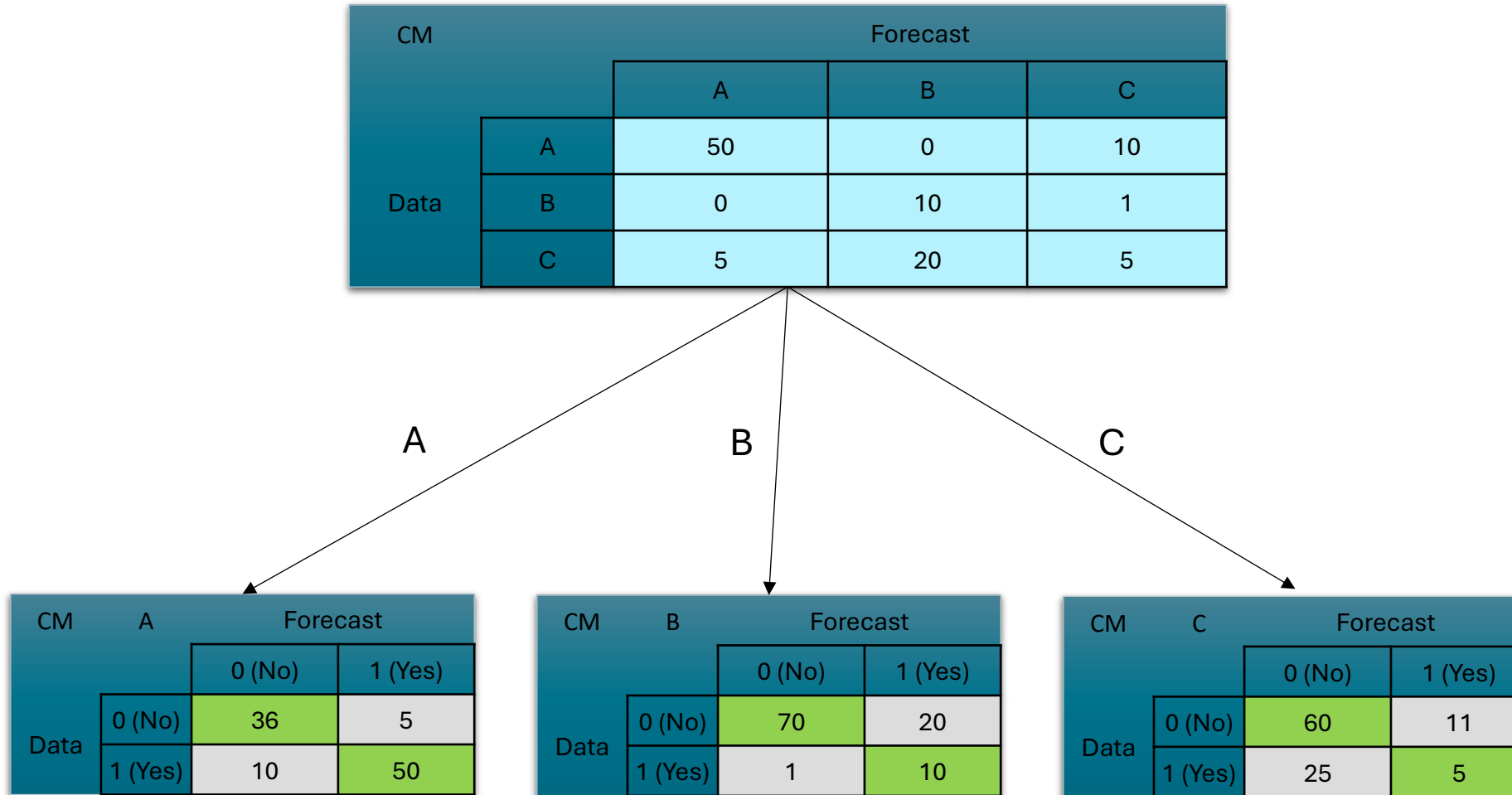
Confusion matrices for non-binary classification tasks

- For multi-class classification the confusion matrix contains one row and one column per class.
- Basic idea: consider each class as a positive class and the others as negative classes, and aggregate the resulting quality measures
- Example (see next slides for details of how to use it):

Conf. Mat.		Forecast		
Data		A	B	C
	A	50	0	10
	B	0	10	1
	C	5	20	5



EVALUATING MULTI-CLASS CLASSIFICATION MODELS



But this gives us multiple scores? How to compute an overall quality score?

EVALUATING MULTI-CLASS CLASSIFICATION MODELS – MICRO VS. MACRO AVERAGING

There are two options for averaging the desired quality measure per class:

- **Micro averaging:** Sums the categories FN, TP and FP across classes and inserts these sums into the definition

$$P = \frac{TP_A + TP_B + TP_C}{TP_A + TP_B + TP_C + FP_A + FP_B + FP_C}$$

$$R = \frac{TP_A + TP_B + TP_C}{FN_A + FN_B + FN_C + TP_A + TP_B + TP_C}$$

- **Macro averaging:** Computes the quality measure per class and averages it

$$P = \frac{1}{3}(P_A + P_B + P_C) \quad R = \frac{1}{3}(R_A + R_B + R_C)$$

- Consequently, for calculating Micro and Macro F1 scores:
 - Micro: Use micro-averaged precision and recall and compute the harmonic mean
 - Macro: Compute the F1 score per class and average these scores

Examples
assume
a three-class
problem
(A, B, C)!

Which of these two is better in case of class imbalance?

EXAMPLE: MICRO VS. MACRO AVERAGING

CM	A	Forecast	
		0 (No)	1 (Yes)
Data	0 (No)	36	5
	1 (Yes)	10	50

CM	B	Forecast	
		0 (No)	1 (Yes)
Data	0 (No)	70	20
	1 (Yes)	1	10

CM	C	Forecast	
		0 (No)	1 (Yes)
Data	0 (No)	60	11
	1 (Yes)	25	5

Micro

Macro

$$P_A = \frac{50}{50 + 5} = \frac{50}{55}$$

$$P_B = \frac{10}{10 + 20} = \frac{10}{30}$$

$$P_C = \frac{5}{5 + 11} = \frac{5}{16}$$

$$R_A = \frac{50}{10 + 50} = \frac{50}{60}$$

$$R_B = \frac{10}{1 + 10} = \frac{10}{11}$$

$$R_C = \frac{5}{25 + 5} = \frac{5}{30}$$

$$P = \frac{50_A + 10_B + 5_C}{50_A + 10_B + 5_C + 5_A + 20_B + 11_C} = \frac{65}{101} \approx 0.64$$

$$R = \frac{50_A + 10_B + 5_C}{50 + 10_B + 5_C + 10_A + 1_B + 25_C} = \frac{65}{101} \approx 0.64$$

$$P = \frac{1}{3} \left(\frac{50}{55} + \frac{10}{30} + \frac{5}{16} \right) \approx 0.52$$

$$R = \frac{1}{3} \left(\frac{50}{60} + \frac{10}{11} + \frac{5}{30} \right) \approx 0.64$$

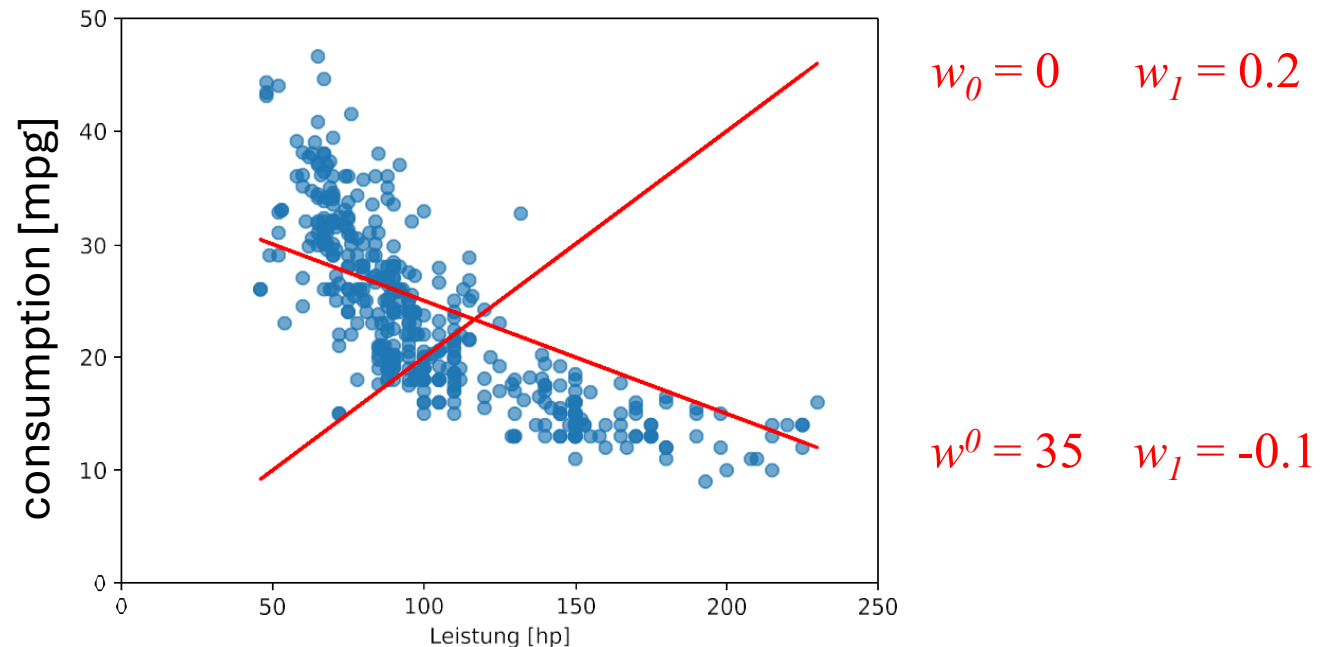
CLASSIFICATION: WHICH METRIC SHOULD I CHOOSE?

Key: **tp** = True Positive, **tn** = True Negative, **fp** = False Positive, **fn** = False Negative

Metric Name	Metric Formula	Code	When to use (and when not)
Accuracy	$\text{Accuracy} = \frac{tp + tn}{tp + tn + fp + fn}$	tf.keras.metrics.Accuracy() or sklearn.metrics.accuracy_score()	+ if easy interpretation necessary - unbalanced data - If FPs and FN have different costs -> You can introduce weights!
Precision	$\text{Precision} = \frac{tp}{tp + fp}$	tf.keras.metrics.Precision() or sklearn.metrics.precision_score()	+ if FPs are more relevant - if FNs are more relevant (since not event considered!)
Recall	$\text{Recall} = \frac{tp}{tp + fn}$	tf.keras.metrics.Recall() or sklearn.metrics.recall_score()	+ if FNs are more relevant - if FPs are more relevant (since not event considered!)
F1-score	$\text{F1-score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	sklearn.metrics.f1_score()	+ if you need a combined view - if you need to be sure what kind of error happens more often
Confusion matrix	NA	Custom function or sklearn.metrics.confusion_matrix()	+ to see which classes are mixed up more often - for estimating overall quality

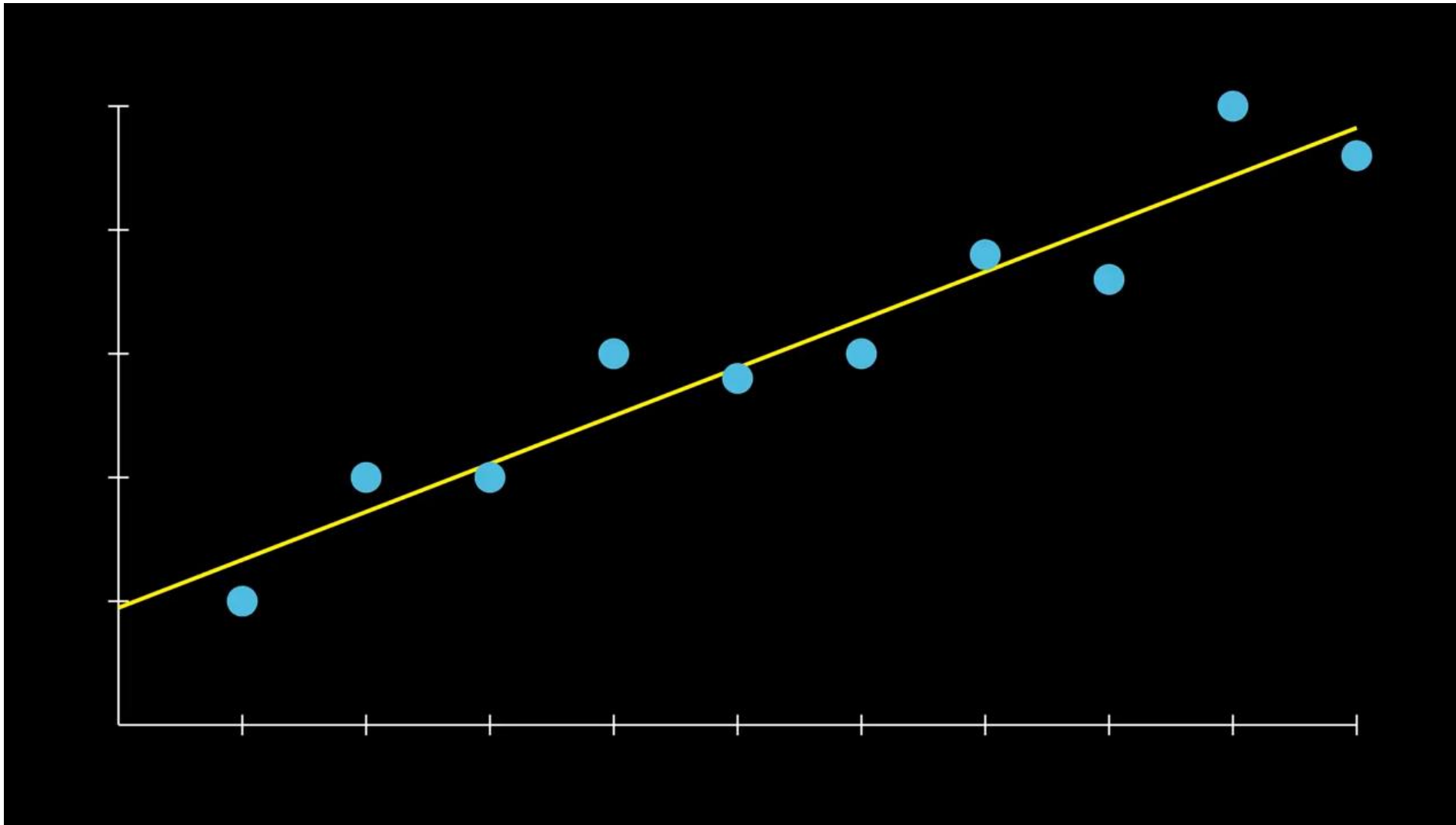
EVALUATING REGRESSION MODELS – A CLOSER LOOK

- Let us assume that we have a linear model with two parameters that can be trained: w_0 and w_1
- Thus, our model is basically a linear function and training means finding the offset and the orientation of “the straight line”



- But what line is the best? We need a quality criterion!
- Example for such a quality criterion: A loss function that is **the sum of squared errors** (see video on the next slide)

EVALUATING REGRESSION MODELS – SSE LOSS



There are other loss functions
(see metrics part in
this slide set)

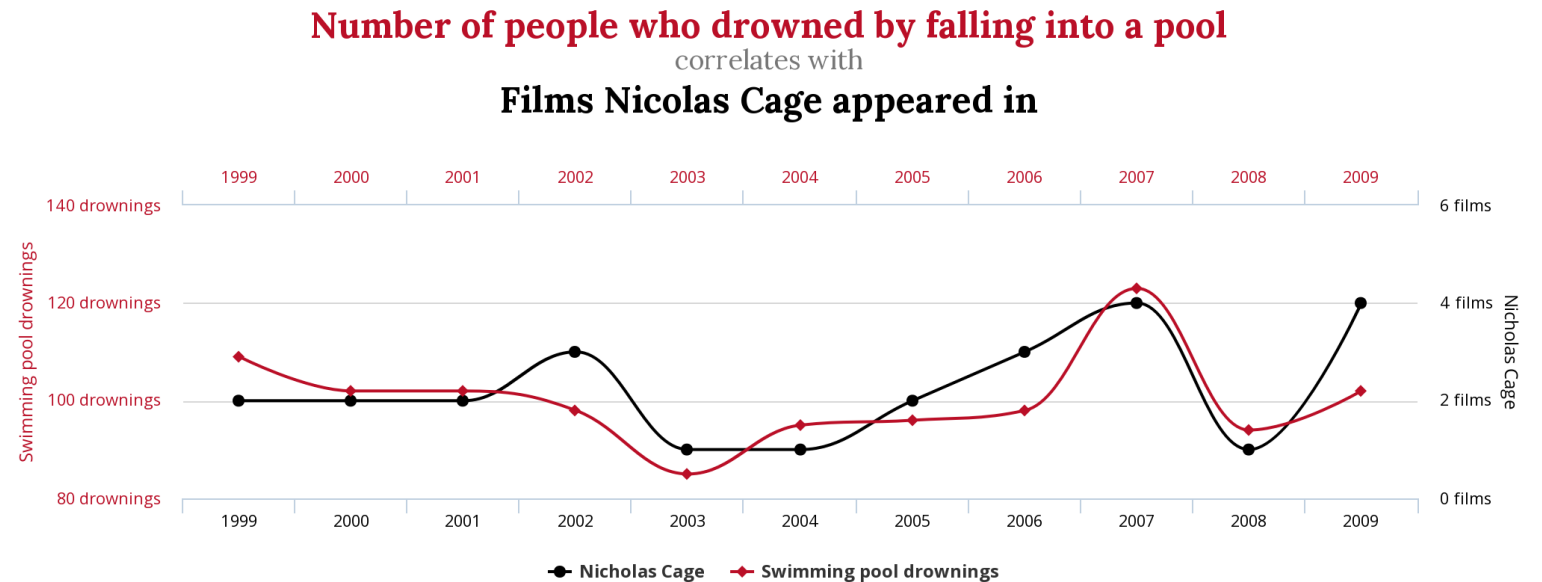
Full version: https://www.youtube.com/watch?v=3dhcmeOTZ_Q

CHASING PHANTOMS: CORRELATION VS. CAUSALITY

- **Correlation** between two characteristics does not mean that there is also causality.

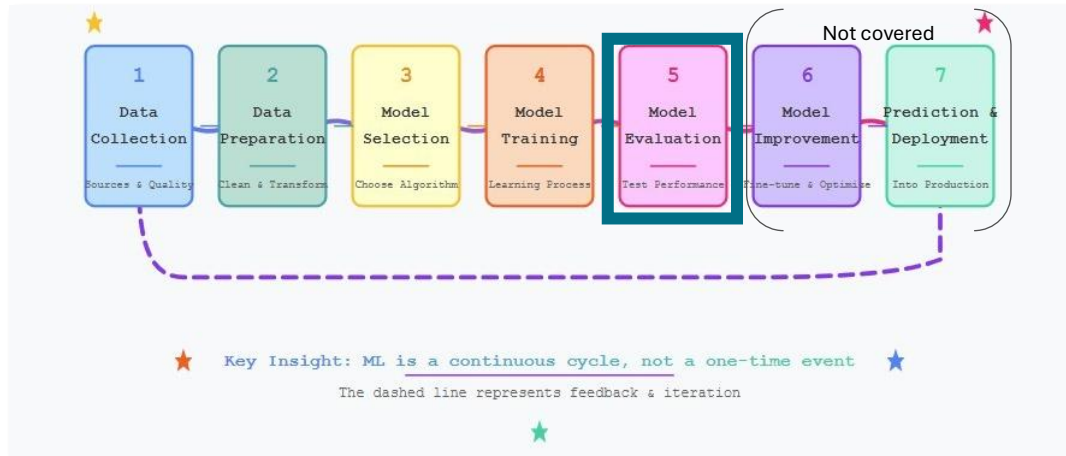
- Example:

- Correlation of the day:
<http://www.correlated.org>
- Spurious Correlations:
<http://www.tylervigen.com/>



tylervigen.com

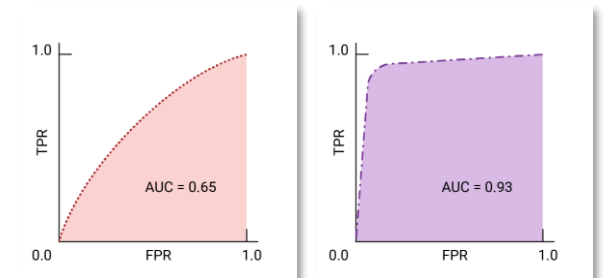
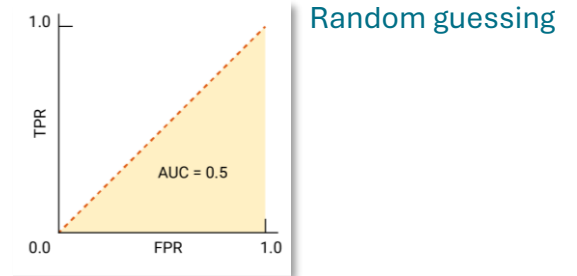
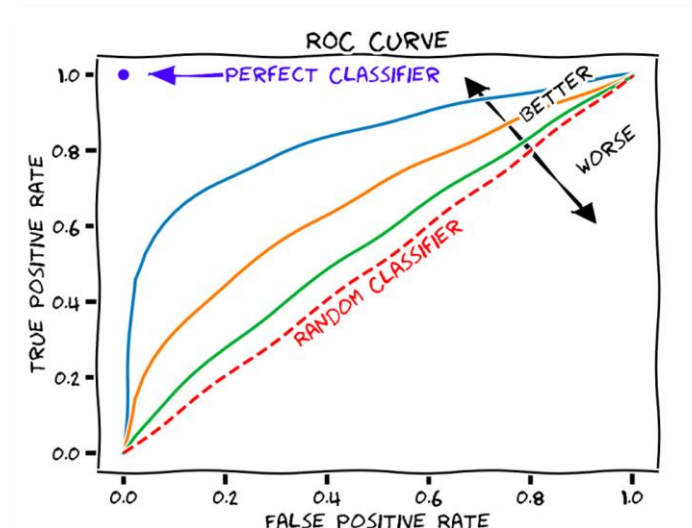
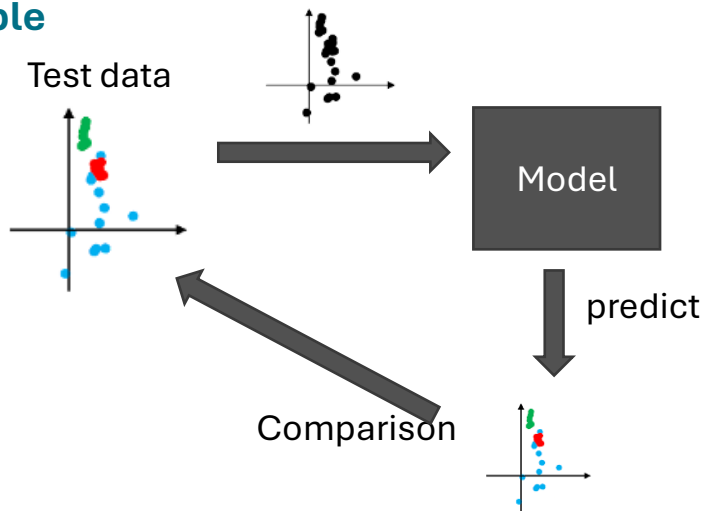
A MACHINE LEARNING PROCESS – MODEL EVALUATION



Testing the trained model on unseen test data (

- Measures how well the model generalizes, i.e., it estimates how well the model will perform in production contexts
 - > metrics can be the same like in the training stage
 - > however, insights are different in the model evaluation stage!
- Another task: Testing the effects of decision boundaries
 - In a 2-class problem the decision boundary is 0.5 by default
 - However, we can modify the boundary if we require more confidence
 - In multi-class problem we can again apply micro or macro averaging

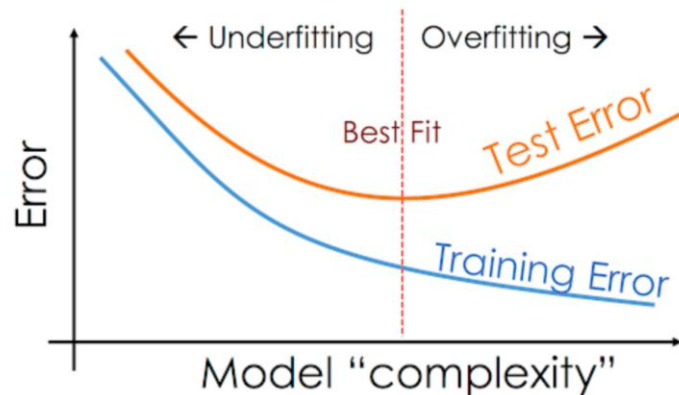
Principle



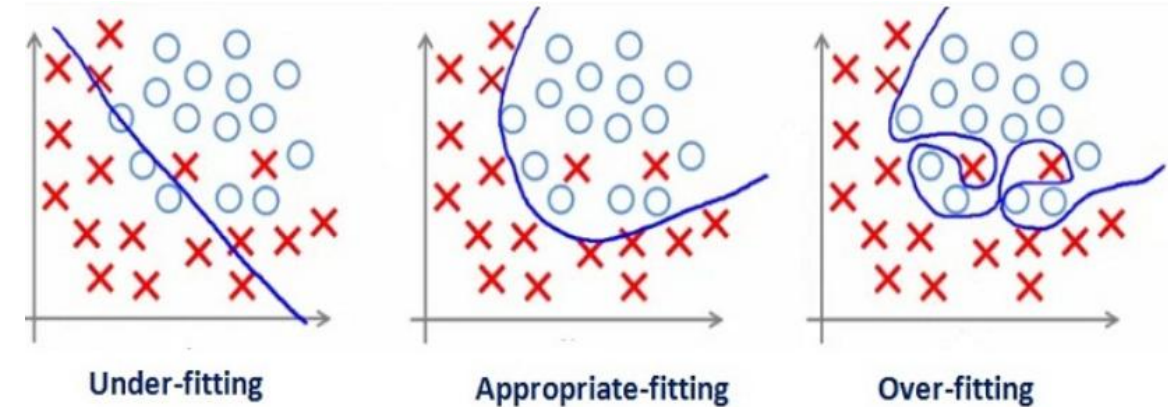
Models better than random guessing

OVERFITTING/UNDERFITTING OF CLASSIFICATION AND REGRESSION MODELS

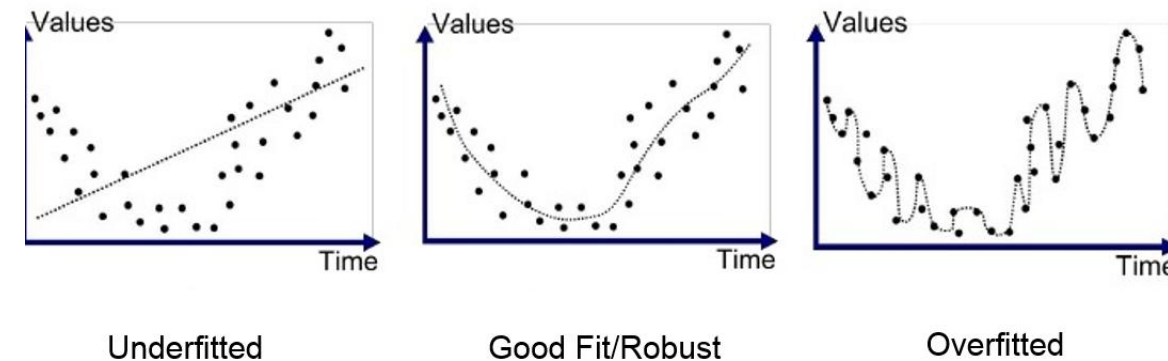
- Error high in training and test: underfitting
- Error low in training and significantly higher in test: overfitting
- Overfitting by:
 - Model complexity too high
 - Too many training epochs
 - Lack of data (variety, overall amount, ...)



Classification



Classification



QUESTIONS?





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